

Lam Fung Academy

SSPA Mock Exam (1) · Primary 5 Mathematics

Scope: Lessons 1–10 (Multiple Figures • Estimation • Area • Fractions)

Exam time: 50 minutes Full score: 100 points

Name: _____

date: _____

Fraction: _____ / 100

Part A: Multiple choice questions

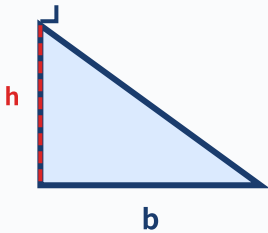
Part A - Multiple Choice Questions

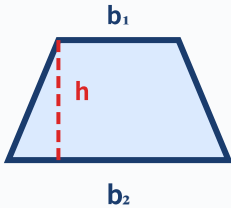
10 questions × 3 points = 30 points

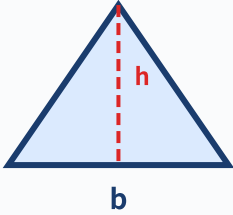
Instructions: Choose the correct answer and fill in the letter of the answer in the space to the right of each question. There is only one correct answer for each question.

Exam tips (SSPA must read)

- ① Do the questions you know first, don't get stuck on one!
- ② Check the unit for each question (cm vs cm² vs cm³)!
- ③ Geometry questions: If there is an elevation in the picture, use the height. If there is no elevation, use the formula to find it!
- ④ Application question: Write an answer sentence! Points will be deducted for not writing steps!
- ⑤ 5 minutes left: Check if the MC has been filled in and if the unit is correct!

question number	Question	fraction
1	What is the Arabic numeral for "seven million and five hundred"? A. 7,000,500 B. 7,500,000 C. 7,050,000 D. 7,005,000	___ / 3 points
2	Round 38,752 to the nearest thousand and use the approximate value: $38,752 \times 4 \approx ?$ A. 120,000 B. 156,000 C. 160,000 D. 155,008	___ / 3 points
3	▲ Triangle: base 12cm, height 8cm  Triangle base length 12cm, height 8cm. Area = ?	___ / 3 points

	A. 96 cm^2 B. 48 cm^2 C. 40 cm^2 D. 20 cm^2	
4	Compare $\frac{2}{3}$ and $\frac{3}{5}$, which one is correct? A. Directly compare the numerators, $3 > 2$, so $\frac{3}{5}$ is larger B. After common division $\frac{10}{15} < \frac{9}{15}$, so $\frac{2}{3}$ is smaller C. After common division $\frac{10}{15} > \frac{9}{15}$, so $\frac{2}{3}$ is larger D. The two fractions are equal	___/ 3 points
5	▲ Trapezoid: Pay attention to height $\neq$ hypotenuse!  Trapezoid: upper base 5 cm, lower base 9 cm, hypotenuse 7 cm, height 4 cm. Which of the following is the correct way to find area? A. $(5 + 9 + 7) \times 4 \div 2$ B. $(5 + 9) \times 7 \div 2$ C. $(5 + 9) \times 4 \div 2$ D. $5 \times 9 \times 4 \div 2$	___/ 3 points
6	Calculate $\frac{1}{2} + \frac{1}{3} = ?$ Which of the following calculations is correct? A. $\frac{1+1}{2+3} = \frac{2}{5}$ B. $\text{LCM}(2,3)=6 \rightarrow \frac{3}{6} + \frac{2}{6} = \frac{5}{6}$ C. $1 + 1 = 2, 2 + 3 = 5$, answer $= 2\frac{1}{5}$ D. $\frac{1}{2} + \frac{1}{3} = \frac{2}{6}$	___/ 3 points

question number	Question	fraction
7	▲ Parallelogram: base 2m, height 50cm (note the unit!)  The parallelogram has a base 2m long and a height 50cm. Area = ? (Tip: Pay attention to the units!) A. 100 cm² B. 1 m² C. 10,000 cm² D. 1,000 cm²	___/ 3 points
8	How many "0"s are there in the number 5,060,402 that need to be read? A. 0 pieces B. 1 piece C. 2 pieces D. 3 pieces	___/ 3 points
9	Calculate $\frac{5}{6} - \frac{1}{4} = ?$ Which step is the right first step? A. $5 - 1 = 4$, $6 - 4 = 2 \rightarrow \frac{4}{2} = 2$ B. $\text{LCM}(6,4)=12$, $\frac{5}{6} \rightarrow \frac{10}{12}$, $\frac{1}{4} \rightarrow \frac{3}{12}$ C. $\text{LCM}(6,4)=24$, $\frac{5}{6} \rightarrow \frac{20}{24}$ D. Direct subtraction: $\frac{5-1}{6} = \frac{4}{6}$	___/ 3 points
10	The rectangle is 8 cm long and 5 cm wide, ▲ Rectangle 8 × 5cm • Dig out a 3 × 3cm square  Dig out a square with a side length of 3 cm in the middle. Remaining area = ? A. 40 cm² B. 31 cm² C. 49 cm² D. 37 cm²	___/ 3 points

Part B: Short answer questions

Part B - Short Questions/Calculation Questions

8 questions × 5 points = 40 points

Instructions: Write the complete calculation steps and answers in the answer area below each question. Answers must be reduced (if applicable).

question number	Question	fraction
1	Write the Chinese pronunciation of the number 3,050,708, and write down the digit where "3" is located and its value. 	___ / 5 points
2	Approximate 4,856,729 to: (a) Ten thousand digits (b) Hundred thousand digits 	___ / 5 points
3	The parallelogram has a base length of 15 cm and a height of 8 cm. Area = ? cm ² 	___ / 5 points
4	The base of the triangle is 20 cm long and the height is 9 cm. Area = ? cm ² 	___ / 5 points

question number	Question	fraction
5	Trapezoid: upper base 6 cm, lower base 14 cm, height 5 cm. Area = ? cm ²	___/ 5 points
6	Calculation: $\frac{2}{3} + \frac{3}{4} = ?$ Write down the steps for common division and reduction.	___/ 5 points
7	Calculation: $\frac{7}{8} - \frac{1}{3} = ?$ Write down the steps for common division and reduction.	___/ 5 points
8	Calculation: $12 \times \frac{3}{4} = ?$ Write down the calculation process and answer with the simplest fraction (or whole number/mixed number).	___/ 5 points

Part C - Application Problems3 questions $\times$ 10 points = 30 points

Instructions: Each question must list all calculation steps and write a complete answer. The answer must be accompanied by the unit.

Question 1 (10 points) – Area of a combined figure**____ / 10 points**

The picture below is made up of a rectangle and a triangle.

Rectangle: length 10 cm, width 6 cm.

Triangle: base 10 cm, height 5 cm, spliced on top of the rectangle.

- (a) Find the area of the rectangle. (2 points)
- (b) Find the area of the triangle. (3 points)
- (c) Find the total area of the entire combined figure. (1 point)
- (d) If the designer scales up the entire figure by a factor of 2 (length of each side $\times$ 2), what is the new total area? (4 points)

Answer area - please write all calculation steps and complete answer sentences:

Question 2 (10 points) – Multi-step application of fractions**____ / 10 points**

Xiao Ming has a box of building blocks with a total of 120 blocks.

in $\frac{1}{3}$ It's red, $\frac{1}{4}$ is blue and the rest is yellow.

- (a) How many red blocks are there? (2 points)
- (b) How many blue blocks are there? (2 points)
- (c) What fraction of the total number of red and blue bricks are there? (2 points)
- (d) How many yellow blocks are there? What fraction of the total? (4 points)

Answer area - please write all calculation steps and complete answer sentences:

Question 3 (10 points) – Comprehensive trap question (estimate + area)**____ / 10 points**

A rectangular playground is approximately 98 m long (rounded to the nearest metre) and 61 m wide (rounded to the nearest metre).

- (a) Use an estimation strategy to approximate the length and width to the nearest ten to estimate the playground area. (3 points)
- (b) Calculate the exact area of the playground using actual measurements. (2 points)
- (c) How many m^2 does the estimated area differ from the exact area? (2 points)
- (d) If 8 students can stand for morning exercises per 10 m^2 , how many students can stand at most using the exact area? (3 points)

Answer area - please write all calculation steps and complete answer sentences:

— Complete the entire volume —

Total score: _____ / 100 points

Signature of marking teacher: _____ Date: _____

Exam tips:

- Part A MC Trap: Question 3: Note that the triangle must be $\div 2$. Question 5: The hypotenuse is not high.
Question 7: Unify the units first.
- Part B calculation questions: To get the general score, you must first find LCM, and the answer must be simplified
- Part C application questions: Each question has its own trap (combined graph segmentation, residual fraction benchmark, estimation vs. exact comparison)
- Make good use of 50 minutes: 12 minutes for Part A $\rightarrow$ 20 minutes for Part B $\rightarrow$ 15 minutes for Part C $\rightarrow$ 3 minutes for inspection

Appendix: Simulation 1 Post-examination Supplementary Practice

24 questions · For after-class consolidation

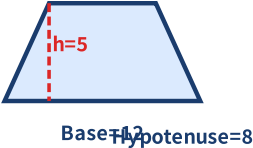
Instructions: The following questions are designed to target common denominator-losing traps in Simulation 1. Please complete them after the exam to consolidate what you have learned.

more number of digits basics (total 6 questions)

#	Question	Difficulty
S1	Write the number: eighty-three million and six hundred and twenty =? -----	
S2	Read out the Chinese pronunciation of 74,030,005 (indicate which "zeros" need to be read and which do not). -----	
S3	In 5,080,604,002, where are "5" and "8" respectively? How much do they represent? -----	
S4	Use the numbers 2, 0, 0, 8, 0, 0, 5, 1 to form the largest eight-digit number and the smallest eight-digit number.	

S5	Approximate 8,054,792 to (a) tens of thousands (b) millions.	
S6	Compare sizes: 7,654,321, 7,645,312, 7,654,231 in descending order.	

areatrap (total 6 question)

#	Question	Difficulty
S7	Triangular base 18 cm, height 7 cm. Area = ?	
S8	The upper base of the trapezoid is 6 cm, the lower base is 14 cm, the hypotenuse is 9 cm, and the height is 5 cm. Which data is not needed? Area = ?	
S9	The area of a parallelogram is 72 cm^2 and its base is 12 cm. High = ?	
S10	The large rectangle is $12 \times 9 \text{ cm}$, and a small rectangle of $5 \times 3 \text{ cm}$ is dug out. Remaining area = ?	
S11	Square perimeter 40 cm. Area = ?	
S12	 Base=12 Hypotenuse=8 The parallelogram has a base of 12 cm, a hypotenuse of 8 cm, and a height of 5 cm. Area = ? Perimeter = ? (Note: area is related to height, perimeter is related to hypotenuse)	

fractioncalculate (total 6 question)

#	Question	Difficulty
S13	$\frac{2}{5} + \frac{1}{3} = ?$ Write down the general steps.	
S14	$\frac{5}{6} - \frac{3}{8} = ?$	
S15	$\frac{3}{4} + \frac{5}{6} - \frac{1}{3} = ?$	
S16	Comparing $\frac{3}{4}$ and $\frac{5}{7}$, which one is larger?	
S17	There are 60 candies in a box, $\frac{2}{5}$ is strawberry flavor, $\frac{1}{3}$ is mango flavor, and the rest are mint flavor. How many peppermint flavors are there?	
S18	Xiao Ming has \$120. He uses $\frac{1}{4}$ to buy books, and then uses $\frac{1}{3}$ under his body to buy stationery. How much money does he have left?	

applicationquestioncomprehensive (total 6 question)

#	Question	Difficulty
S19	The upper base of the trapezoidal garden bed is 5 m, the lower base is 9 m, and the height is 4 m. If 4 flowers are planted per m^2 , how many flowers will be planted in total?	
S20	The population of Hong Kong is approximately 7,346,000 (rounded to the nearest thousand). Between what two numbers might the actual population lie?	

S21	Rectangle $2.5\text{ m} \times 180\text{ cm}$, area = ? cm^2 (unify the units first)	
S22	A rectangle and a triangle both have an area of 36 cm^2 . The length of a rectangle is 9 cm. How wide is it? The base of a triangle is 12 cm, what is its height?	
S23	A trapezoid, the upper base is $\frac{1}{3}$ of the lower base, the lower base is 12 cm, and the height is 6 cm. Area = ?	
S24	The combined figure consists of a triangle (base 10 cm, height 6 cm) and a rectangle ($10\text{ cm} \times 8\text{ cm}$). Total area = ? If each side of the entire figure is enlarged by a factor of 2, how many times is the new area?	

 Post-exam review record:

Part A score: ____/30 Part B score: ____/40 Part C score: ____/30 Total score: ____/100

The question types that need to be strengthened most in this exam: _____; The traps that need to be reviewed: _____

Lam Fung Academy • LF Academy • We don't teach math. We teach trap avoidance.